Fine-tuning Comparison Report (GreenTrainer2)

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1. Inputs

Green (carbon-aware):
/content/fair/green_carbon/green_20260619_035153/green_run_report_20260619_035153.json

Baseline:
/content/fair/baseline/baseline_20260619_034636/baseline_run_report_20260619_034636.json

2. Disaggregated Results

Metric	Green (carbon-aware)	Baseline	Impact (A vs B)
Total Energy	3.0658 Wh	3.3615 Wh	-9.6%
GPU Energy	2.4630 Wh	2.6667 Wh	-8.3%
CPU/SoC Energy	0.6028 Wh	0.6947 Wh	-15.3%
Avg GPU Power (W)	69.63	64.38	+7.5%
Avg CPU/SoC Power (W)	10.80	13.00	-20.4%
Time (wall)	138.2 s	145.7 s	-5.4%
CO_2 (operational×PUE)	183.95 mg	201.69 mg	-9.6%
CO_2 (embodied amortized)	0	0	N/A
CO_2 total (PUE+embodied)	183.95 mg	201.69 mg	-9.6%

3. GreenTrainer2 Efficiency Metrics

Metric	Green (carbon-aware)	Baseline	Impact (A vs B)
Skipped steps	69	0	+100.0%
Skip rate (%)	21.37%	0.00%	+100.0%
Early accum exits	15	0	+100.0%
Micro-batches saved	87	0	+100.0%

4. Visualizations

Figure 1: Total Energy (Wh)

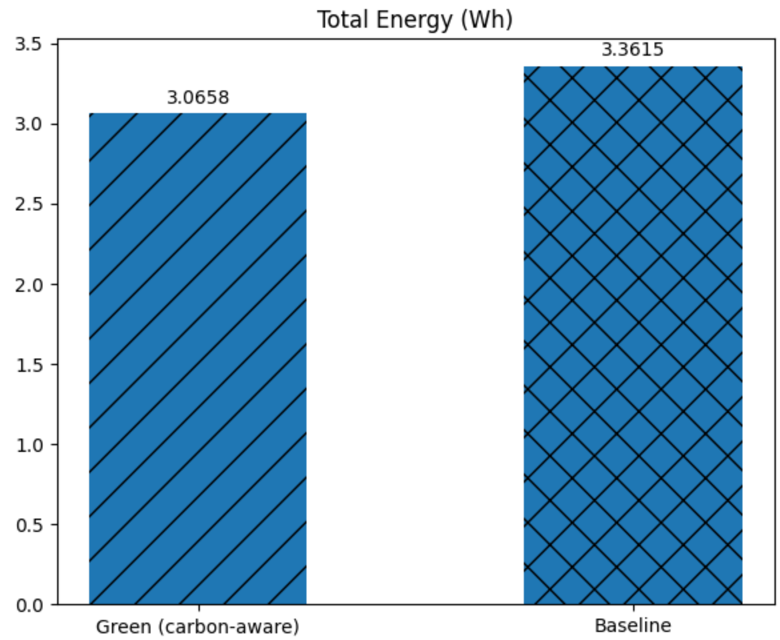


Figure 2: Training Time (s)

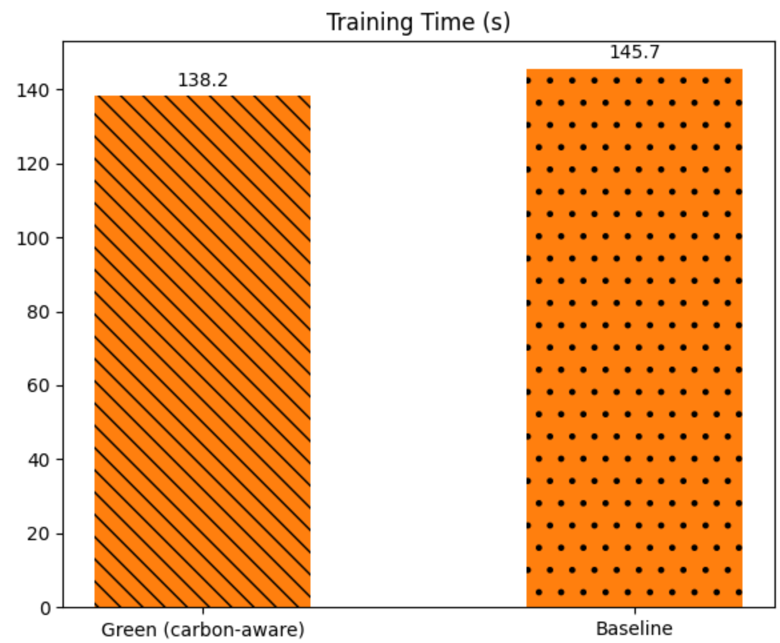


Figure 3: CO_2 Emissions (kg, operational×PUE)

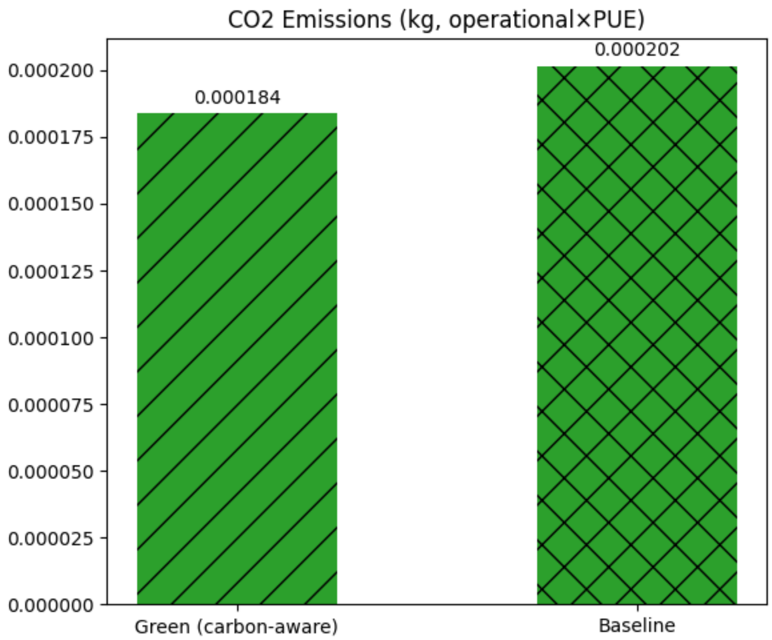
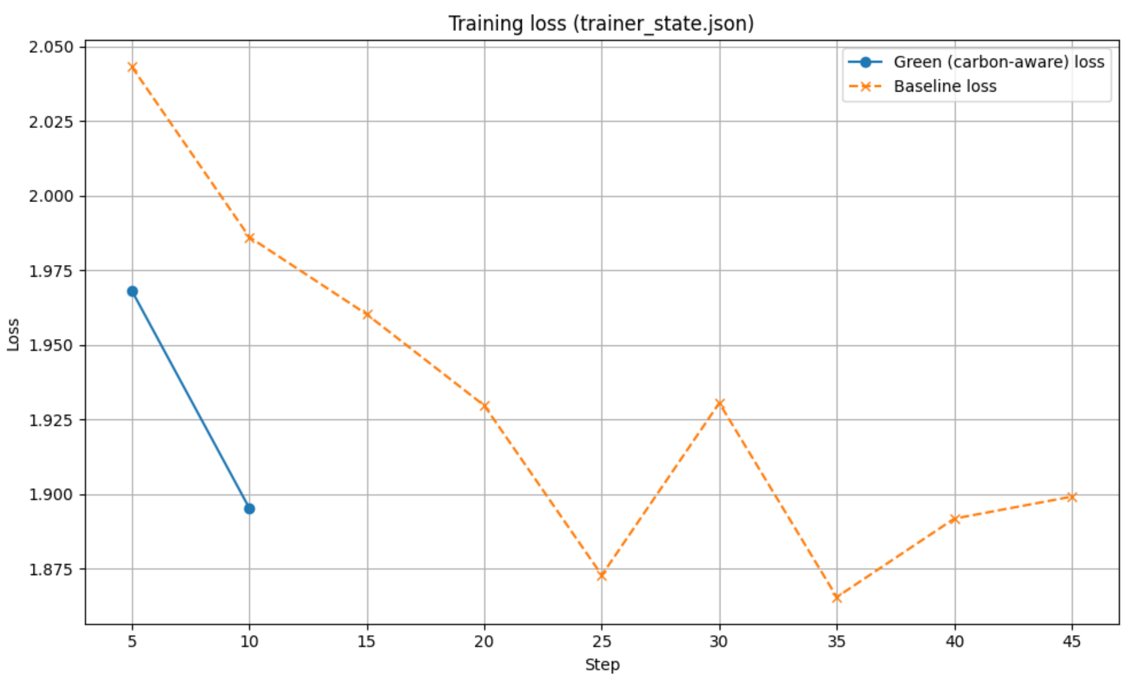


Figure 4: Loss curve



5. Notes

This report is disaggregated: energy, carbon intensity, and PUE-adjustment are reported separately, and embodied amortization is optional. GreenTrainer2 efficiency metrics (skipped steps, skip rate, early accum exits, micro-batches saved) are 0 for baseline runs. If real-time power measurement is unavailable, the run report may indicate a TDP proxy method.